

Creature v06 — wiring schematic (net-label form)

Same net name = same wire. Colour groups: red power, dark grey ground, teal I2C, purple I2S, green LED, plum speaker

ESP32-S3-N16R8

IO5	I2S_SCK
IO6	I2S_WS
IO7	I2S_SD
IO15	I2S_BCLK
IO16	I2S_LRC
IO17	I2S_DIN
IO8	SDA
IO9	SCL
IO4	LED_DATA
3V3	3V3
5V	+5V
GND	GND

BH1750 light

VCC	3V3
GND	GND
SCL	SCL
SDA	SDA
ADDR	GND

BME280 env

VIN	3V3
GND	GND
SCL	SCL
SDA	SDA

IMU ICM20689

VCC	3V3
GND	GND
SCL	SCL
SDA	SDA
XDA	n/c
XCL	n/c
AD0	GND
INT	n/c

INMP441 mic

VDD	3V3
GND	GND
L/R	GND
WS	I2S_WS
SCK	I2S_SCK
SD	I2S_SD

MAX98357A amp

VIN	3V3
GND	GND
LRC	I2S_LRC
BCLK	I2S_BCLK
DIN	I2S_DIN
GAIN	n/c
SD	n/c
OUT+	SPK+
OUT-	SPK-

SK6812 strip

+5V	+5V
DIN	LED_DATA
GND	GND

Speaker 4ohm

+	SPK+
-	SPK-